

A.R.M.O.R

AI-Relay Mining Operation Rover

Transforming blind, high-risk mining operations into precision, remote control management through our LiDAR-driven anti-collision command system backed by continuous mesh connectivity



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A.R.M.O.R. eliminates PAMA’s mining inefficiencies by delivering a Physical Digital Twin and end-to-end platform



 Situation

Fleet idle time exceeds **20–30% of operational** hours

Supervisors **lack real-time visibility** without physical presence on-site

Operators rely on manual judgement **no data driven guidance**

Reactive maintenance causes unplanned downtime & cost spikes

 Complication

Every idle hour = fuel burn + zero output, **compounding OPEX daily**

Blind spots & radio delay create safety risks and slow hazard response

Deviation from mine plan goes undetected until end-of-shift reports

No centralized system to dispatch, monitor, or intervene in **real time**

 Solution



 Impact

Operational Efficiency		Safety & Compliance		Financial Return (Optimistic Scenario)	
Idle time ↓ 50%	Fuel efficiency ↑ 21%	Cycle time ↓ 18%	Production ↑ 15%	NPV (3 Yr) Rp 295.35B net	IRR 568.6%
28% → 14%	2.8 → 2.2 L/ton	24.5 → 20 min	850 → 980 ton/hr	ROI (3 Yr) 1203.4%	Payback 2–3 months

Background

One of the economic sectors that is engaged in the **exploration, extraction and processing of valuable mineral** resources from within the Earth.



PAMA
member of **ASTRA**

Company's Profile

PT Pamapersada Nusantara (PAMA) is a mining contractor under PT United Tractors Tbk, providing end-to-end mining services including mine design, exploration, extraction, hauling, barging, and commodity transportation. With **large-scale mining operations**, PAMA relies heavily on **fleet efficiency, operator safety, and real-time operational visibility**, making AI-IoT solutions highly relevant for improving **safety and productivity**.

Scale of Operations

PAMA operates large-scale **mining activities** involving **heavy equipment fleets**, operators, hauling routes, loading points, dumping areas, and continuous production targets. Due to this scale, even **small improvements in fleet efficiency, idle time, fuel consumption, and operator safety** can create significant **operational and financial impact**.

Challenges

Idle time

High Fuel
Consumption

Unstable Cycle
Time

Operator
Fatigue Risk

As a major mining contractor, **PAMA is pressured** to keep **operations efficient** and **cost-competitive**...

...while **commanding** visibility across every unit, every cycle, in **real time**

A. Supply-Side Pressure



Fleet imbalance & idle time
Over-truck / under-truck mismatches create loading bottlenecks; idle time exceeds >20% of operational hrs/shift



Uncontrolled fuel consumption
No granular tracking per unit per activity, OPEX bleeds invisibly across hundreds of daily cycles.



Reactive maintenance culture
No predictive signal → breakdowns are surprises. Unplanned downtime cascades into missed targets.

B. Demand-side Pressure



Zero real-time operator feedback
Operators rely on manual judgement, no in-cab guidance for digging point, route, or dumping location.



Safety & fatigue exposure
High-risk pit areas require physical supervisor presence; no fatigue detection system exists for long-shift operators.



Mine plan deviation undetected
Deviations from short-term mine plan go unreported until end of shift, rework and ore loss accumulate silently.

The cost of inaction vs. what **A.R.M.O.R.** delivers

Metric	Before	After
Idle Time	28%	14%
Fuel (L/ton)	2.8	2.2
Cycle Time	24.5 min	20 min
Production	850 t/hr	980 t/hr

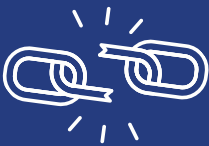
> Rp130B projected annual benefit
Operational savings validated by pilot data

~300+ heavy equipment units at risk of idle daily
Supervisor blind without physical pit presence

13x optimistic benefit-to-cost ratio at full deployment
Pilot recovers cost in under 3 months

Competitors Deliver Enterprise-Level Mining Solutions at Premium Cost

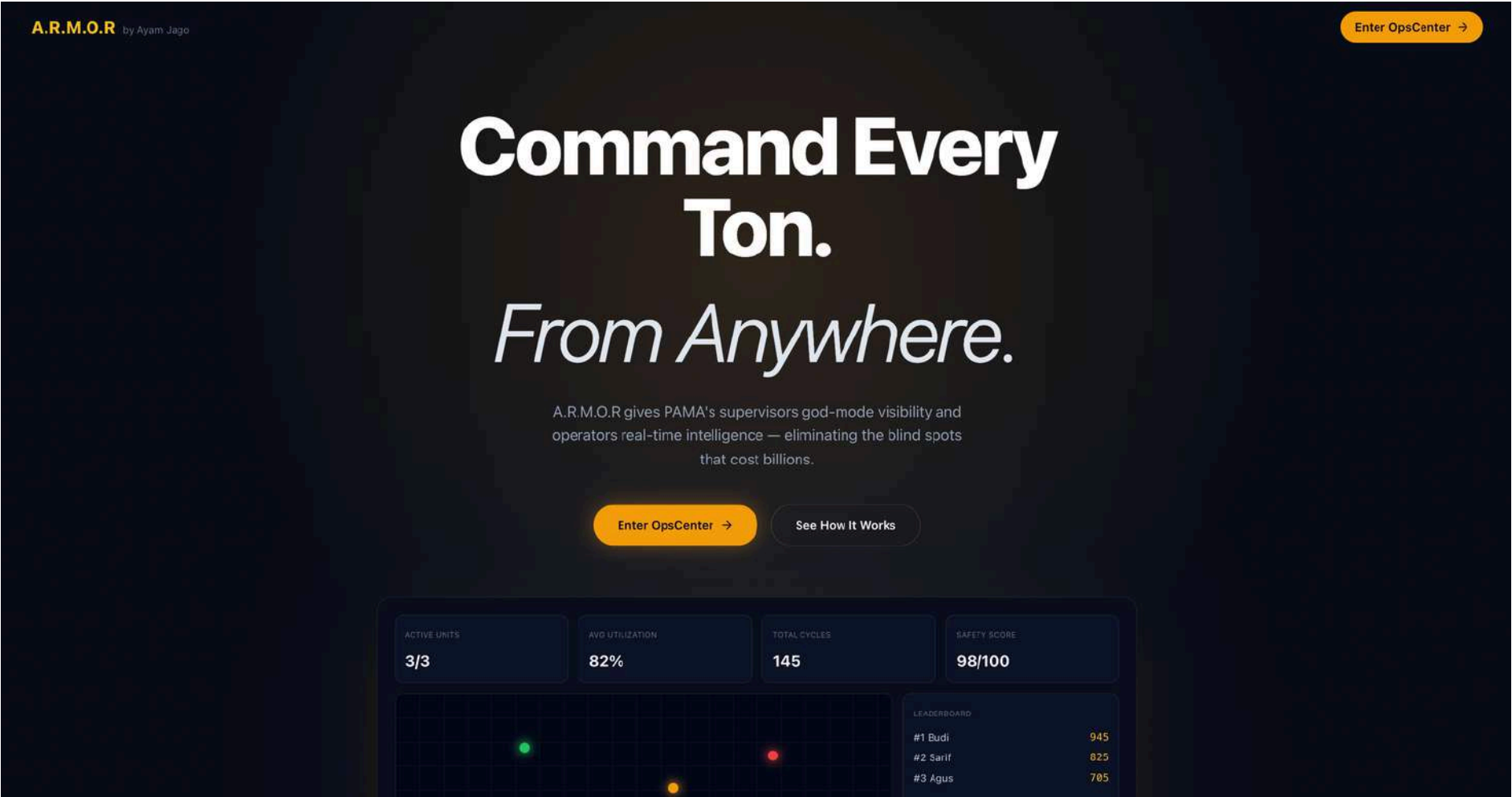


PRODUCT	Caterpillar MineStar	Hexagon Operator Alertness System	Minerva Digital Platform	ControlMaster® Guidance Autonomous Rockbreaker Fleet
 STRENGTH	- Strong enterprise fleet management for real-time tracking - Truck assignment Cycle time - Payload - Productivity monitoring	- Strong mining safety solutions for fatigue detection - Distraction alert Collision avoidance - Proximity warning 360-degree visibility	- In-house AI mining platform using real-time data - AI & big data - ROC Fleet optimization - Fuel & road analysis Dynamic dispatch	- First autonomous rockbreaker fleet used in Grasberg underground - Improves safety by moving operators away from hazardous underground areas Supports remote and autonomous operation from control room
 WEAKNESS	- Expensive Complex - More suitable for large-scale mature operations.	- Safety-focused - Expensive Less suitable for low-cost solution	Not commercially available - High internal investment - Requires strong digital infrastructure and in-house tech capability	Expensive and complex - Requires control room, strong network, and equipment integration - Site-specific, mainly for underground mining

OUR PRODUCT – A.R.M.O.R AI-RELAY MINING OPERATION ROVER

 MAIN FEATURES	Low-cost modular AI-IoT fatigue detection Harsh braking Abnormal vibration Location tracking Risk score Dashboard monitoring	 PRICING	Pilot: Rp2.25B Full Deployment: Rp22.73B
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Snapshots



Description

ARMOR Control System

A.R.M.O.R OpsCenter is a Physical Digital Twin platform that replicates PAMA's Remote Operation Center vision in a deployable web system. Supervisors command the full fleet from anywhere. Operators receive real-time guidance from a safe, connected cockpit. Every unit, every cycle, every anomaly, visible in one place.

Real-Time Fleet Monitoring

Remote Operator Cockpit

AI Decision Support (RAG)

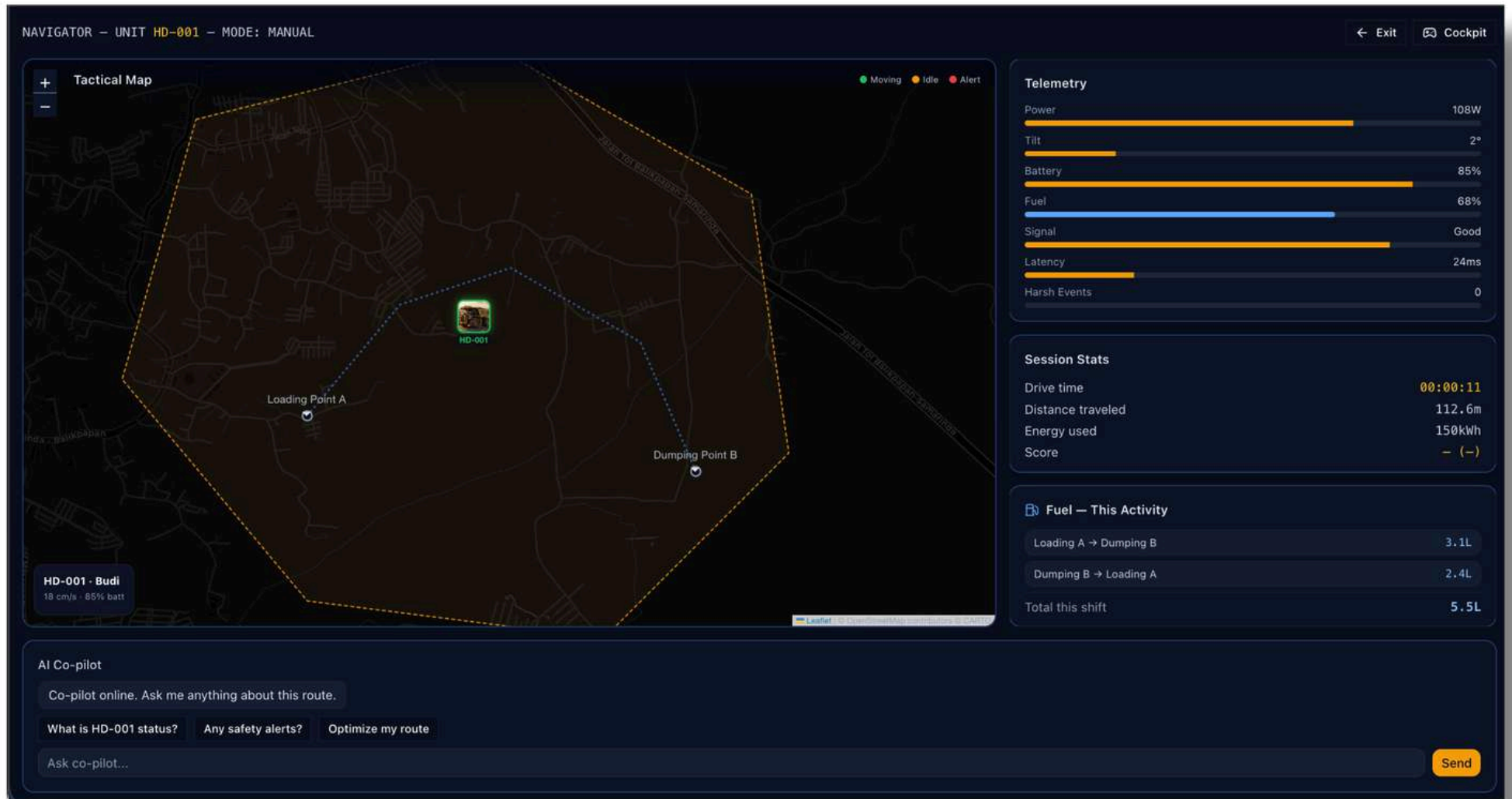
Gamification & Operator Scoring

Link MVP

<https://armor-pamapersada-nusantara.vercel.app/>

Snapshots

Description



Menu Navigator

Fleet Map — Live

Interactive top-down map showing real-time position of all active units, route lines, and zone boundaries updated every 200ms via WebSocket

Operator Leaderboard

Live-ranked scoring board tracking each operator's safety compliance, cycle efficiency, and fuel discipline updated in real time every session

Fuel Monitoring

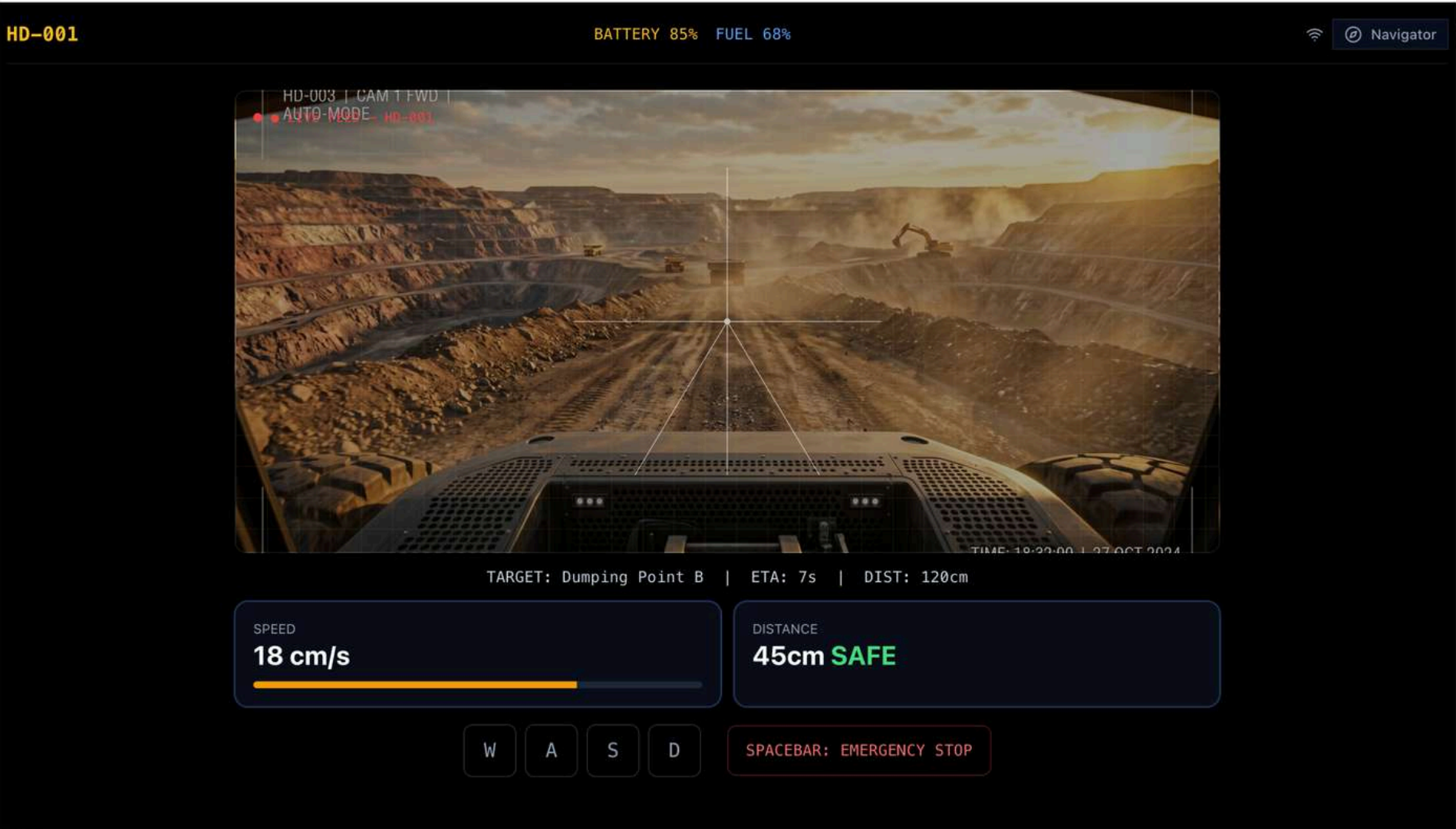
Per-unit fuel consumption tracked across every activity segment, Loading A → Dumping B, with running totals and efficiency flags

AI Chat (ARMOR AI)

RAG-powered assistant answers natural language queries about fleet status, bottlenecks, and anomalies using real-time operational data

Snapshots

Description



Menu Cockpit

Live Camera Feed

Full-screen camera stream from the unit's onboard cameras with crosshair overlay and AUTO MODE indicator for situational awareness

WASD Keyboard Control

Full keyboard drive interface W/A/S/D for movement, Spacebar for emergency stop with real-time key highlight feedback on screen

Speed & Distance HUD

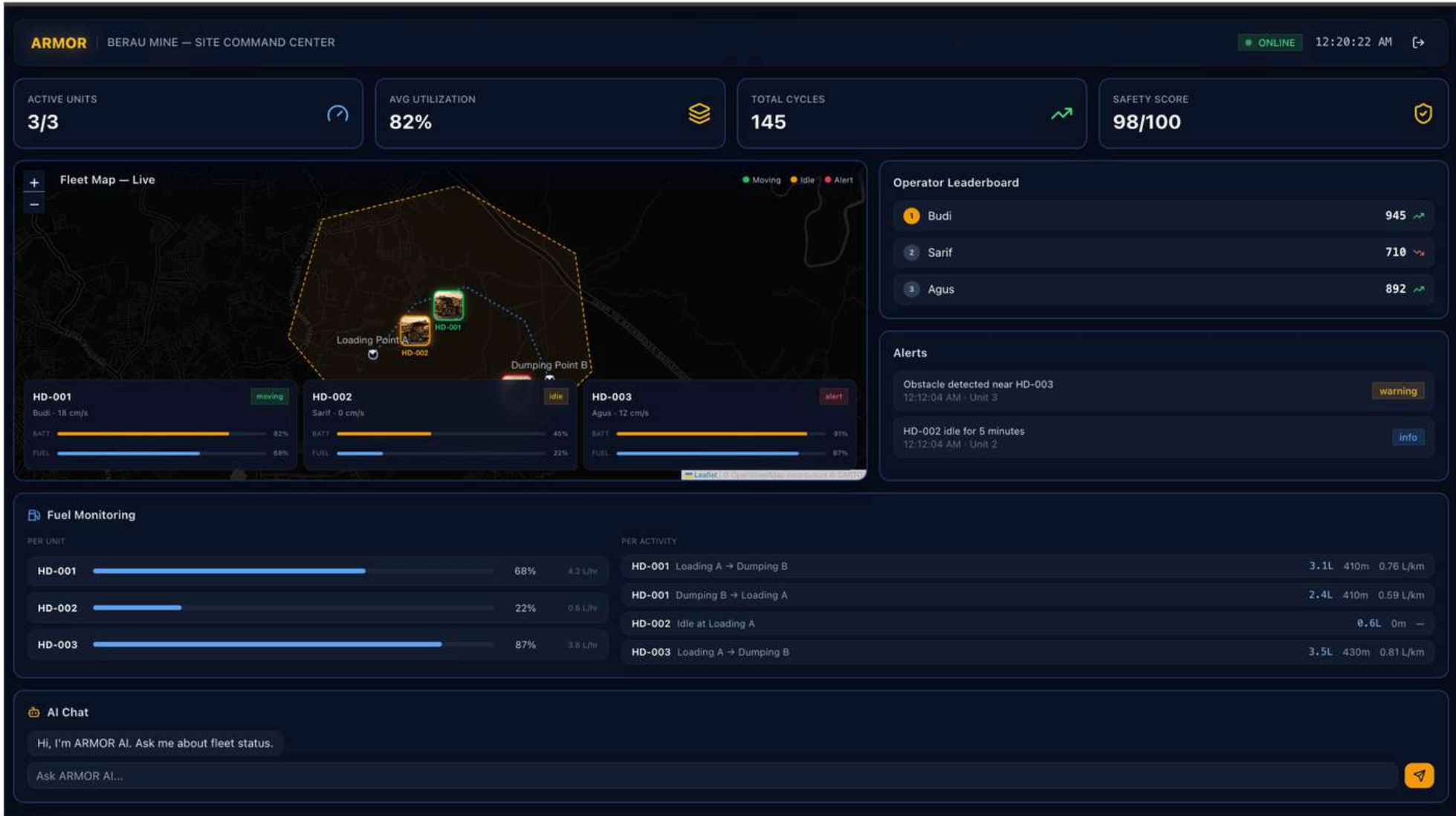
Live telemetry overlay showing current speed (cm/s) and obstacle distance with SAFE/WARNING status derived from LiDAR sensor data

Target Navigation Bar

Persistent mission bar showing current target waypoint, estimated arrival time, and remaining distance synced with the Navigator screen

Snapshots

Description



Menu Supervisor

Tactical Map

Top-down live map of the pit area showing unit position, designated route, Loading Point A and Dumping Point B waypoints with boundary overlay

Telemetry Panel

Real-time sensor readout: Power, Tilt, Battery, Fuel, Signal strength, Latency, and Harsh Event counter all streamed via WebSocket at 200ms

Session Stats & Fuel Log

Live session tracker showing drive time, distance traveled, energy consumed, current score, and per-activity fuel breakdown for the active shift

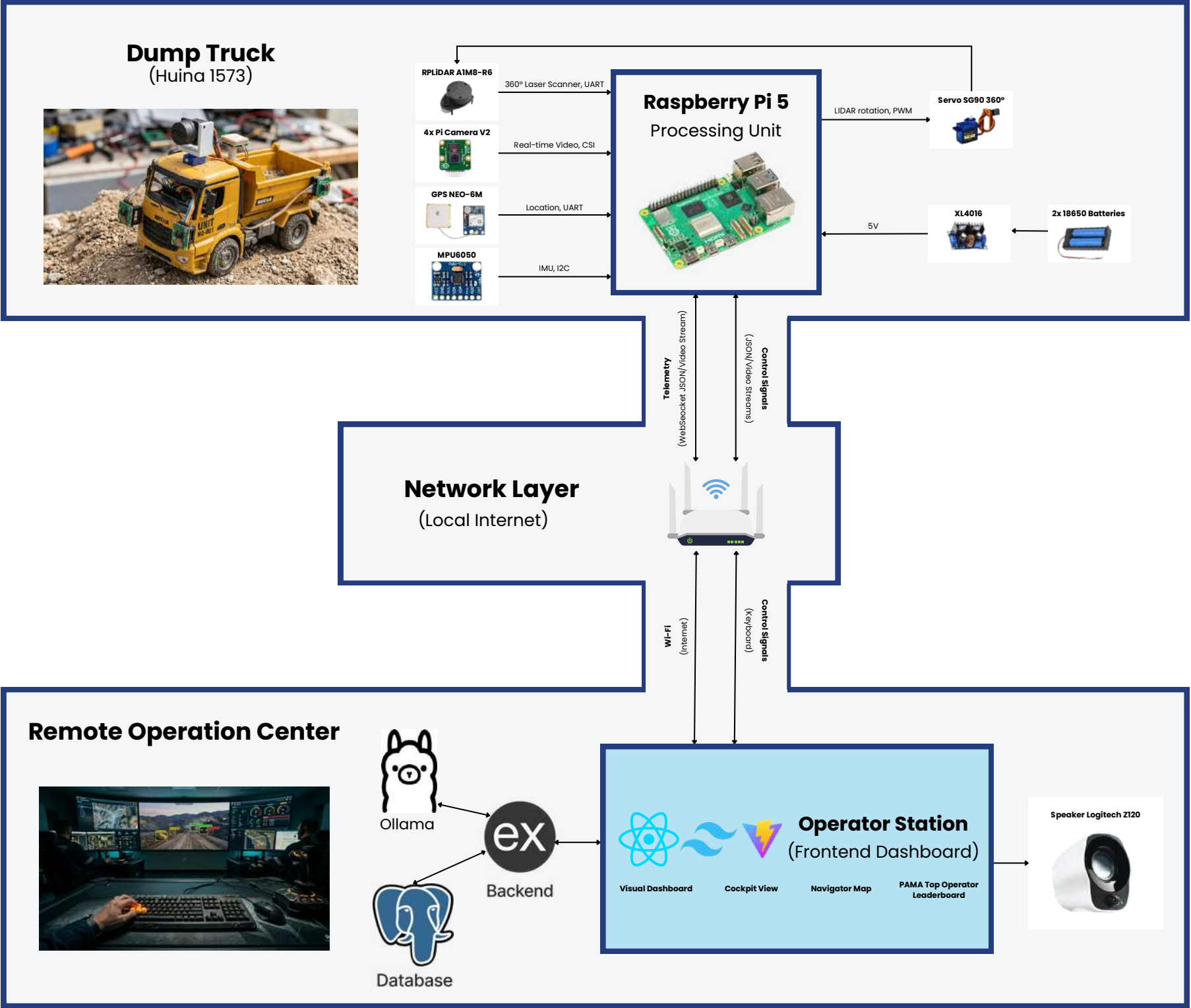
AI Co-Pilot

Context-aware assistant with preset prompts ("What is HD-001 status?", "Any safety alerts?", "Optimize my route") and free-text query support

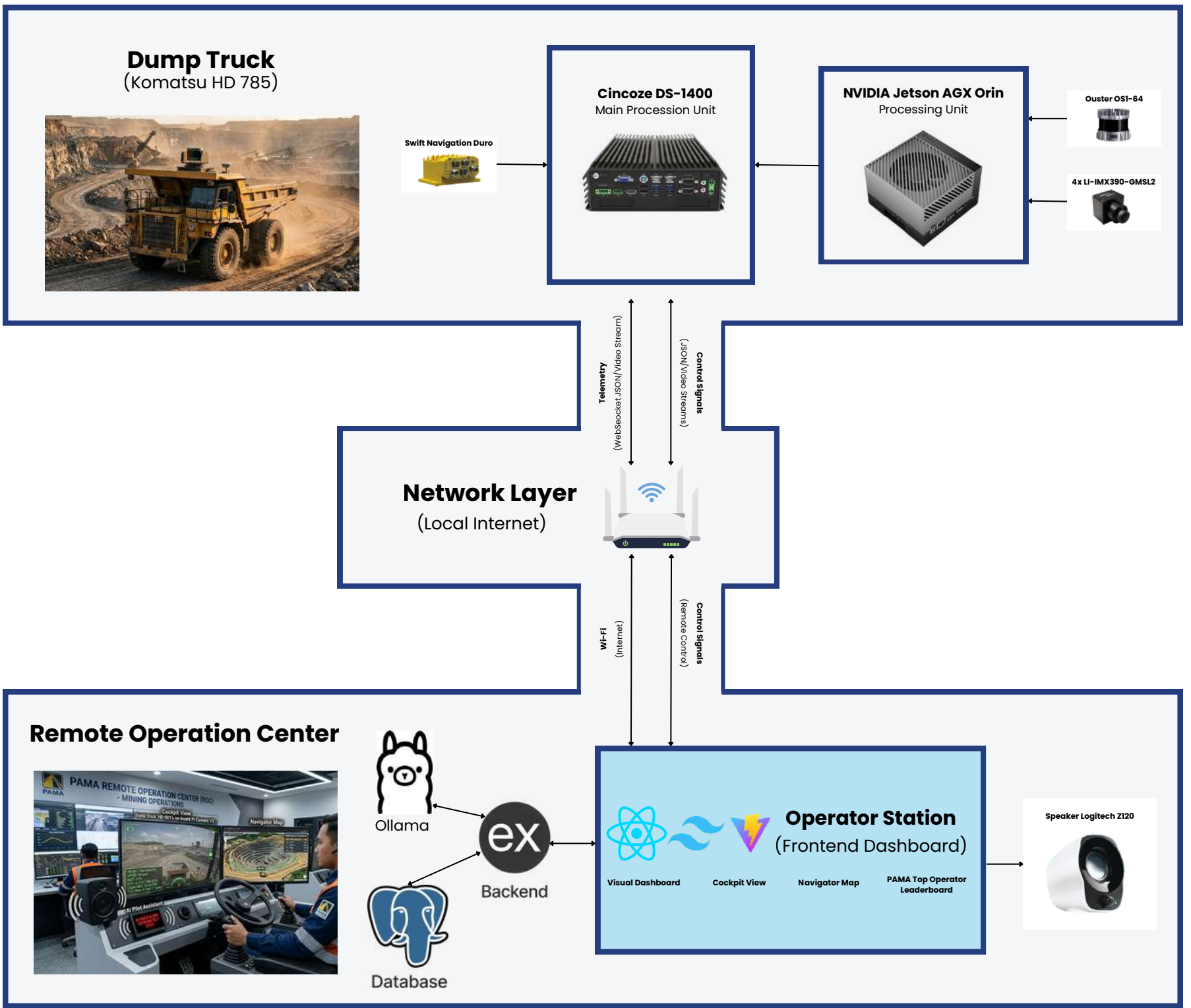
A.R.M.O.R. scales from Raspberry Pi prototype to NVIDIA Jetson-powered ROC at full deployment



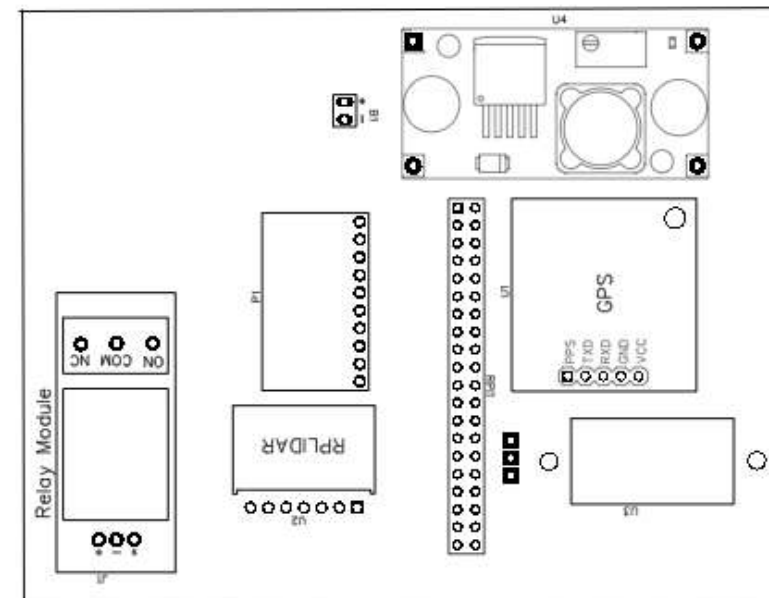
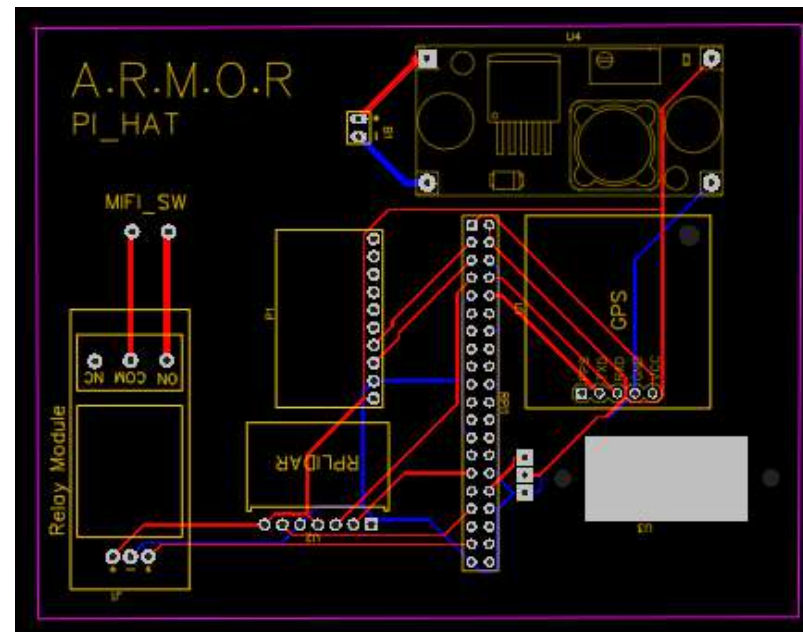
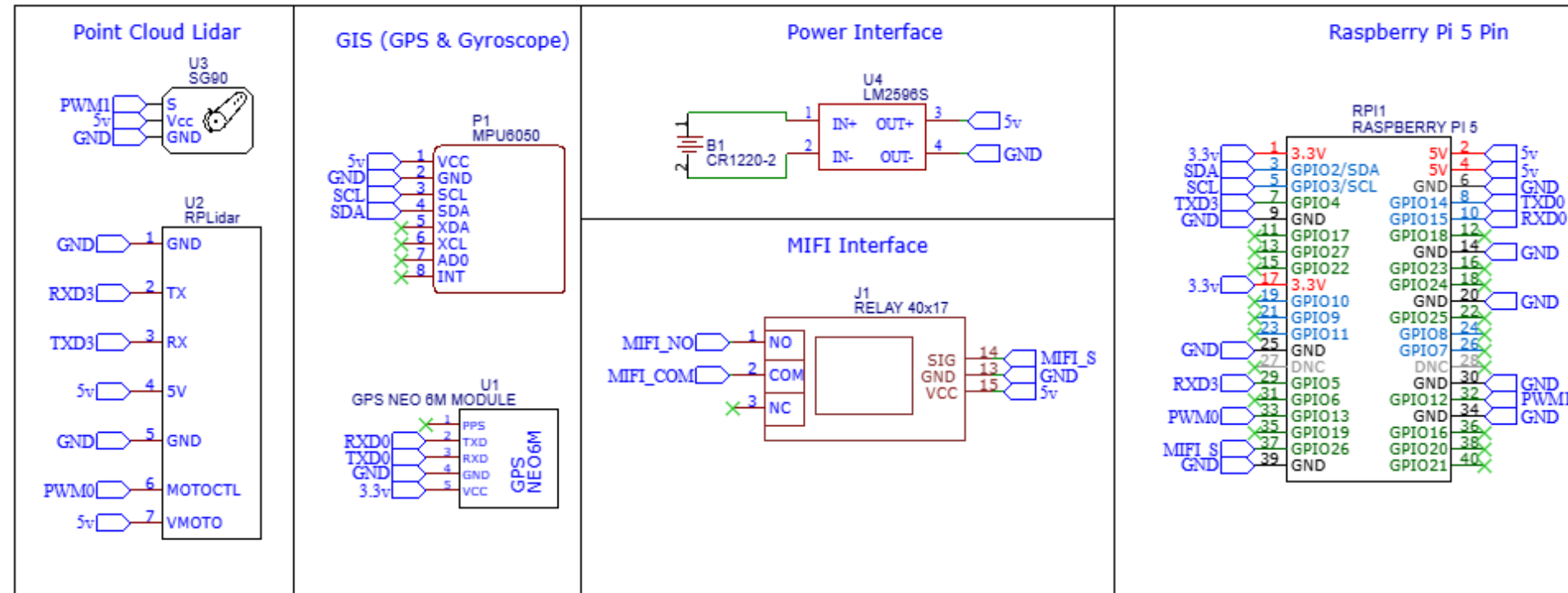
Prototype



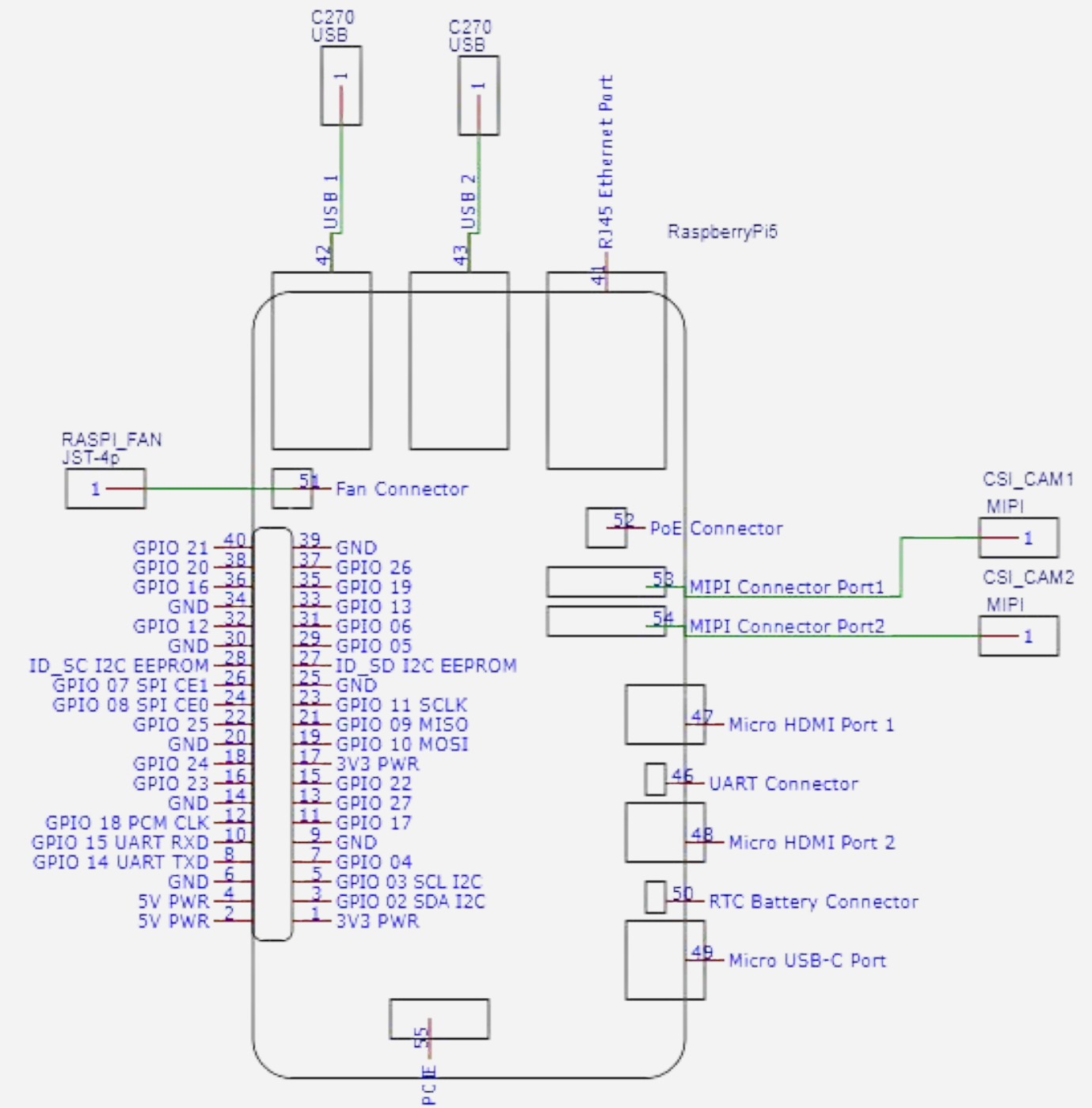
Actual



Raspberry Pi Hat PCB



Raspberry Pi 5 Peripheral Connections



Blank Spot Condition
(Offline)

The Mesh **signal is lost** in the deepest part of the pit

The Raspberry Pi 5 **automatically saves GPS data and temporary AI logs** to a **local database**

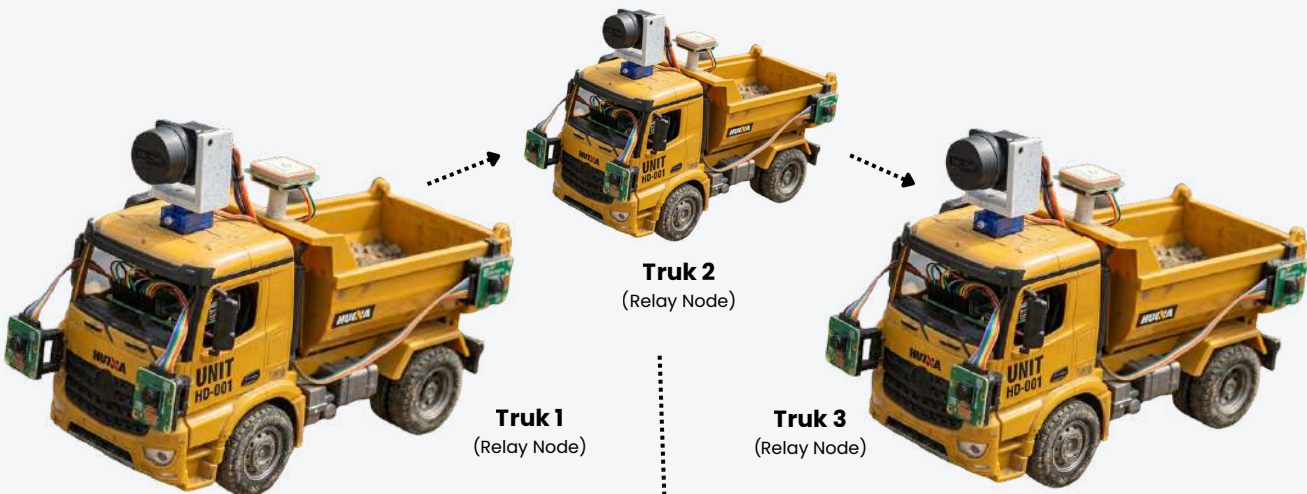
Interconnected Condition
(Online)

The Mesh Repeater **signal has been regained**

The Express.js system automatically **synchronizes pending data to the Central Server** (PostgreSQL on ROC)

Truck Swarm
(Huina 1573)

Wireless Mesh Protocol
(BATMAN)

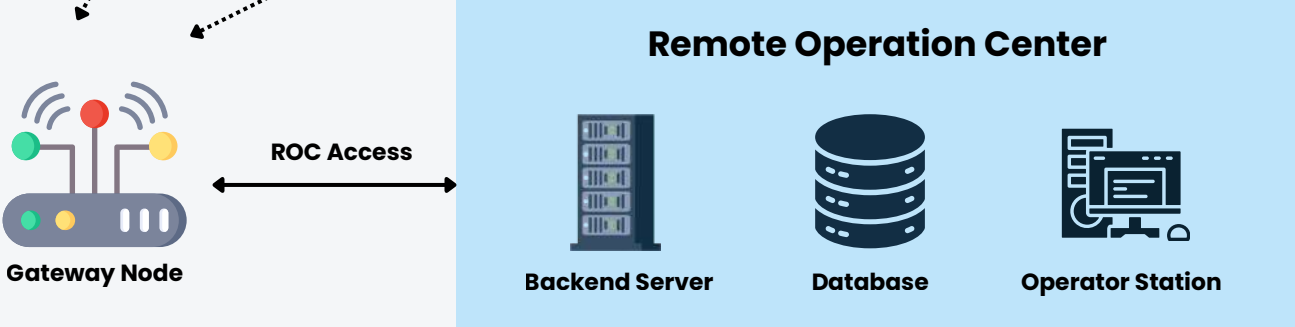


Self-Healing Path
Truck 1 → Truck 2 → Repeater

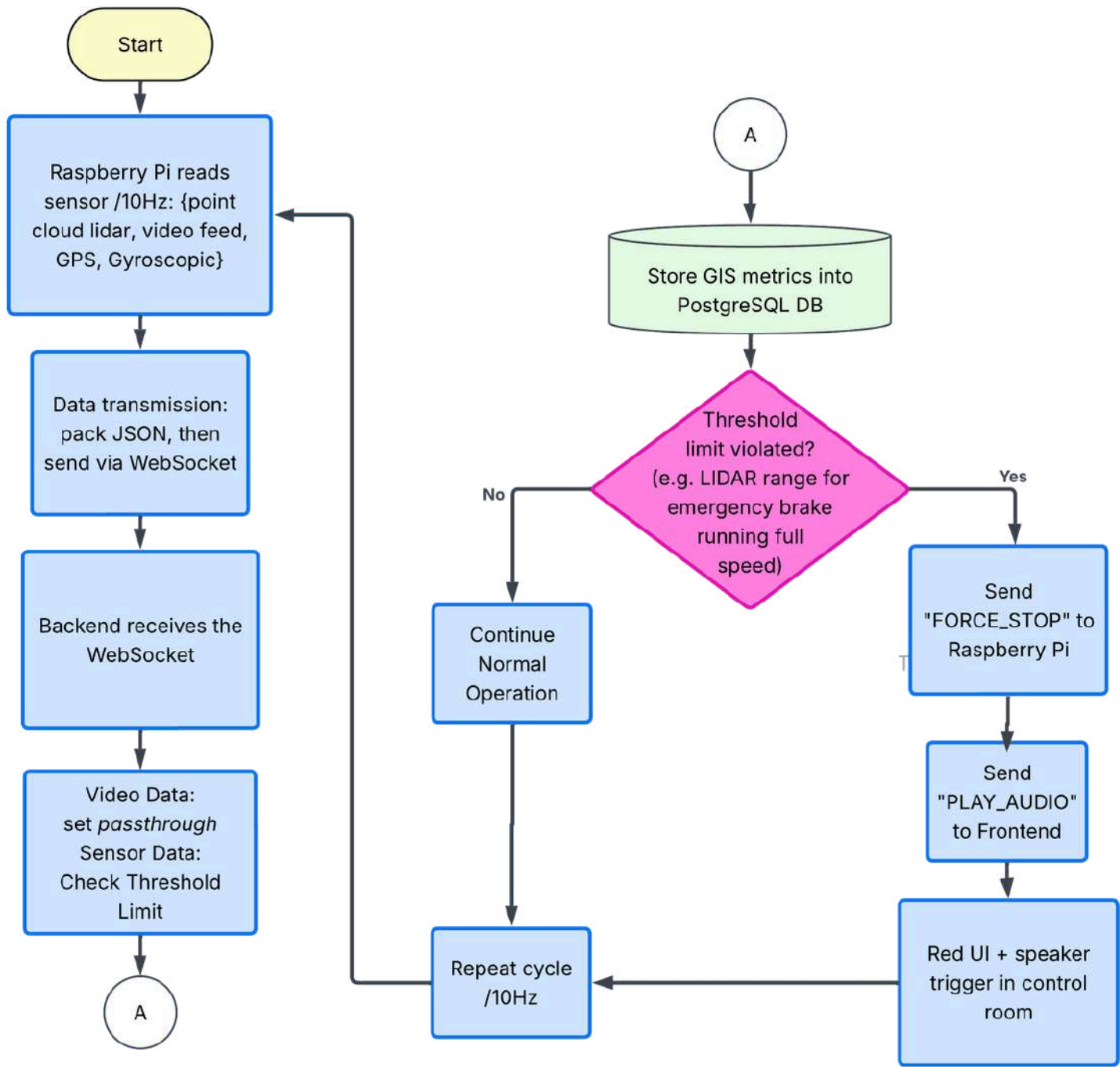
Mesh
Communication
Backbone



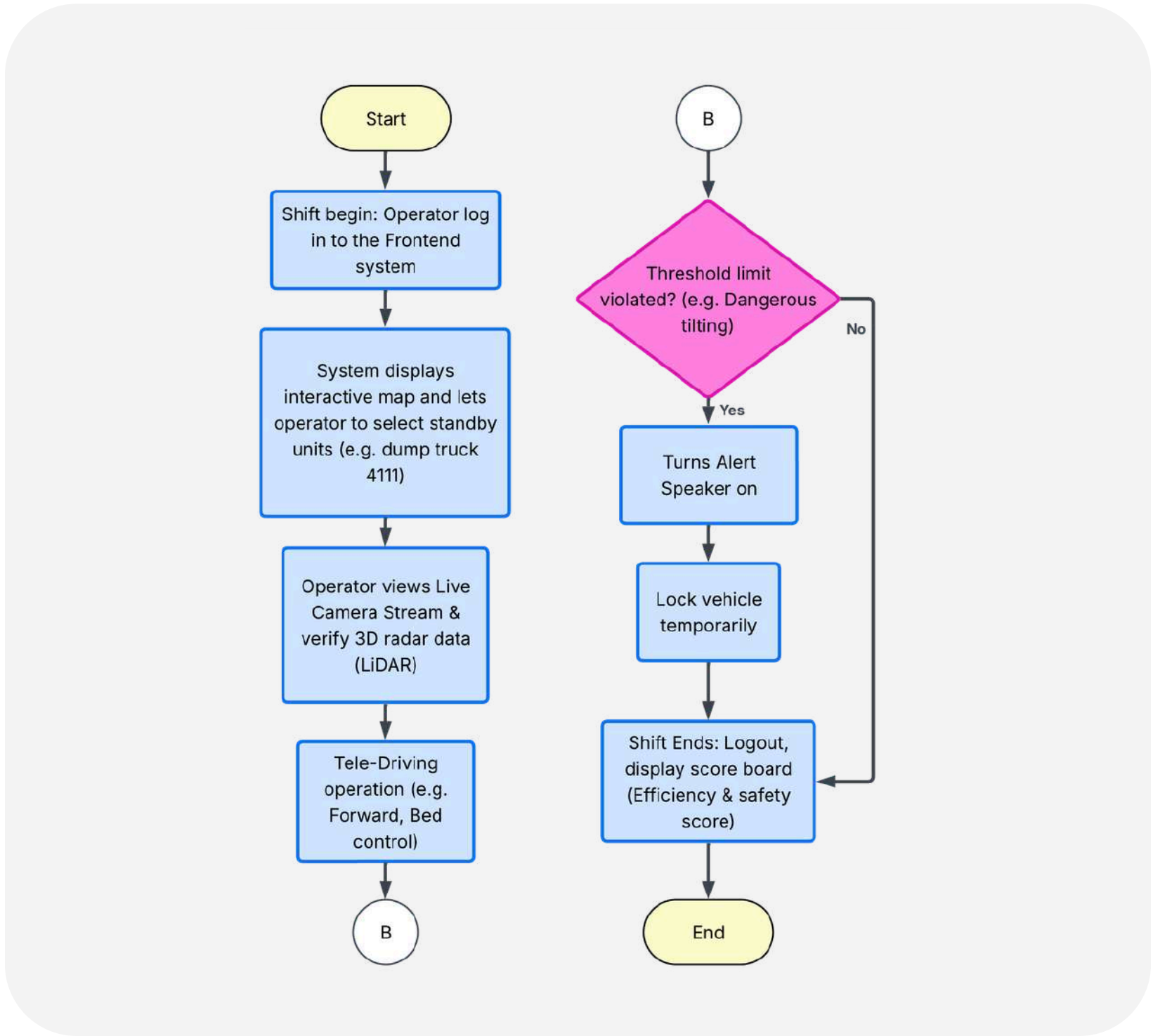
Gateway & ROC
Connection

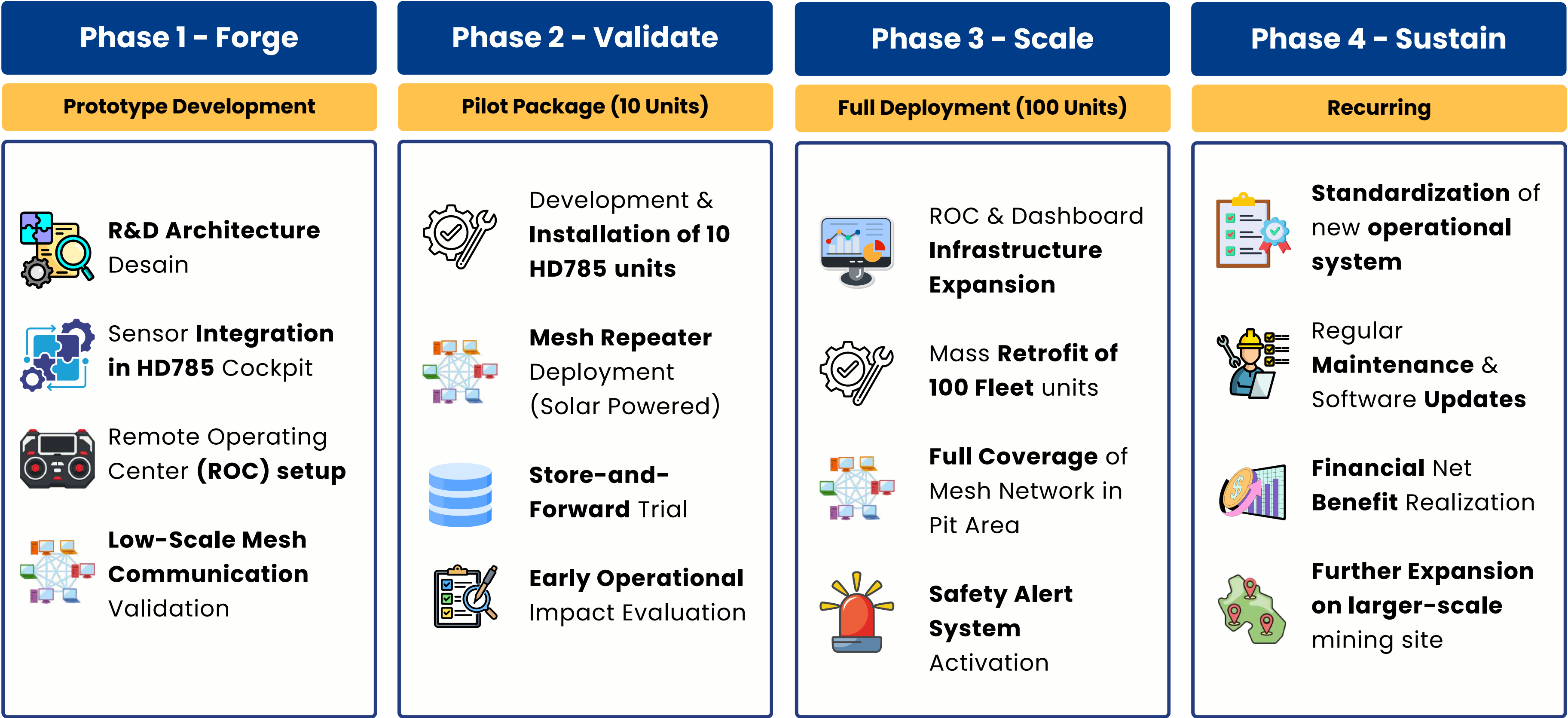


Truck



Operator





Hardware Kit

Components	Price
4x LI-IMX390-GMSL2	Rp 30,900,000
Ouster OS1-64	Rp 50,950,000
Swift Navigation Duro	Rp 50,000,000
NVIDIA Jetson AGX Orin	Rp 36,000,000
Cincoze DS-1400	Rp 27,000,000
Set of Computer	Rp 3,050,000
Total	Rp 197,900,000

Pilot Package (10 units, 3 months)

Components	Calculation	Price
Hardware kit	10 unit * Rp 197,900,000	Rp 1,979,000,000
Installation	10 unit * Rp 3,000,000	Rp 30,000,000
Dashboard setup pilot	Fixed cost	Rp 100,000,000
Training operator & supervisor	Fixed cost	Rp 50,000,000
Software subscription	10 units * Rp 1,200,000* 3 months	Rp 36,000,000
Support pilot	Fixed cost	Rp 50,000,000
Total		Rp 2,245,000,000

Full Site Deployment (100 units, 1 year)

Components	Calculation	Total
Hardware kit	100 units * Rp 197,900,000	Rp 19,790,000,000
Installation	100 units * Rp 3,000,000	Rp 300,000,000
Enterprise Dashboard Setup	Fixed cost	Rp 250,000,000
System Integration	Fixed cost	Rp 600,000,000
Training Operator, Supervisor, HSE	Fixed cost	Rp 150,000,000
Software subscription	100 units * Rp 1,200.000 * 12 months	Rp 1,440,000,000
Maintenance contract	100 units * Rp 2,000,000/yr	Rp 200,000,000
Total		Rp 22,730,000,000

Annual Recurring Cost (for year 2)

Components	Price
Software subscription	Rp 1,440,000,000
Maintenance contract	Rp 200,000,000
Minor replacement & recalibration (10% of hardware kit)	Rp 1,979,000,000
Total	Rp 3,619,000,000

Operational Improvements and Projected Net Benefit

Benefit Scenario Assumptions:

- Assumes 100% of projected operational benefit is achieved
- Monthly financial benefit: Rp10.85B/month
- Annual financial benefit: Rp130.20B/year
- Benefit comes from fuel savings + production improvement
- Represents the highest potential return, subject to pilot validation

Key Operational Metrics

Metric	Before	After 3 months	Improvements
Utilization	68%	83%	15%
Idle Time	28%	14%	-50%
Fuel efficiency	2.8 L/ton	2.2 L/ton	-21%
Cycle Time	24.5 mins	20 mins	-18%
Production	850 ton/hours	980 ton/hours	15%

Financial Impact

Components	Calculation	Value
Production after implementation	980 t/h *24 hrs/day * 30 days	705,600 tn/month
Fuel saving	0.6 L/tn * 705,600 tn/month * Rp 23,000/L	9.75 billion/month
Additional production/month	130 tn/h * 24 h/day * 30 days	93,600 tn/month
Production value	93,600 t/month * Rp 11,700/t	1.1 billion/month

Fuel cost (before) = Rp 39.4 billion/month



PROJECTED FINANCIAL BENEFIT (AFTER IMPROVEMENTS) =
RP 10.85 BILLION/MONTH

TARGET BENEFIT-TO-COST RATIO = 13X

Details of Commercial Offers

PILOT PACKAGE

- Starts with a 3-month pilot on 10 selected units
- Validates system accuracy, user adoption, and operational impact before large investment
- Uses pilot data to refine alert thresholds, dashboard design, and reporting format
- Reduces implementation risk before scaling to 100 units
- After successful pilot, system is expanded into full site deployment
- Full deployment includes enterprise dashboard, API integration, training, maintenance, and site-wide monitoring
- Suitable for PAMA because it allows evidence-based decision-making before committing to full implementation

Rp 2.25 billion

FULL DEPLOYMENT

- Deployment of 100 hardware kits across one mining site
- Full installation, testing, calibration, and system readiness check
- Enterprise dashboard for supervisor, dispatcher, HSE, maintenance, and management
- Real-time monitoring of fatigue, harsh braking, vibration, location, and operator risk score
- Modules included: PAMA Pulse, Operator Benchmark, and Dispatch Optimizer
- API/system integration with existing fleet, HSE, maintenance, or control room systems
- Full user training for operators, supervisors, dispatchers, HSE, maintenance, and admin team
- Maintenance, remote support, software updates, and system health check
- Shift/monthly reporting for operational performance, safety, and efficiency improvement

Rp 22.73 billion

+

CONTINUITY PLAN

- Annual software subscription for SmartCab AI dashboard and analytics
- Maintenance contract for deployed hardware and system support
- Minor replacement for damaged sensors, cameras, mounting, wiring, or small components
- Recalibration of camera, GNSS, LiDAR, and vibration sensors to maintain data accuracy
- Remote technical support and troubleshooting
- Software updates, bug fixes, and feature improvements
- System health check and performance monitoring
- Continuous reporting for fatigue, harsh braking, vibration, operator risk score, and operational KPIs
- Support for dashboard users, including supervisor, dispatcher, HSE, maintenance, and management team
- Ensures the system remains reliable after the first-year deployment

Rp 3.62 billion/year

A.R.M.O.R

AI-Relay Mining Operation Rover

Puts the supervisor in command and
the operator in safety from anywhere

APPENDIX



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KEY PARTNERS - Industrial hardware suppliers (e.g., Swift Navigation, Ouster, Cincoze). - Mesh Network infrastructure vendors (e.g., Rajant or local equivalents). - Mining regulatory bodies (e.g., Ministry of Energy and Mineral Resources / ESDM) for safety compliance and standardization.	KEY ACTIVITIES - R&D for sensor fusion and semi-autonomous driving algorithms. - Hardware retrofitting and physical installation on Komatsu HD785 units. 3D localized mapping of the pit area. - Routine system calibration and predictive maintenance. KEY RESOURCES - Human Capital: Specialized Engineering Team (Robotics, AI/Computer Vision, and Network/IoT). - Intellectual Property: Proprietary navigation algorithms and "Store-and-Forward" delay-tolerant network architecture. - Physical Assets: Local server infrastructure and the Remote Operation Center (ROC).	VALUE PROPOSITIONS - Maximized Productivity: 24/7 continuous hauling operations with zero shift-change or break-time downtime. - Enhanced Safety: Zero safety incidents caused by operator fatigue or human error. - Cost Efficiency: Optimized fuel consumption and reduced tire wear through AI-controlled precision acceleration and braking.	CUSTOMER RELATIONSHIP - Long-term Service Level Agreements (SLA). 24/7 Remote Operation Center (ROC) technical support and monitoring. - Dedicated on-site field engineers during the initial deployment phases. CHANNELS - Direct B2B technical pitches to mining executives and operations managers. - On-site Pilot Project demonstrations (Proof of Concept). - Strategic partnerships with heavy machinery OEMs (Original Equipment Manufacturers).	CUSTOMER SEGMENT - Large-scale coal and mineral mining companies (Tier 1 & Tier 2). - Mining Contractors (focusing on earthmoving and hauling operations).
COST STRUCTURE - Hardware procurement (LiDAR, IP69K Cameras, GNSS, Industrial PCs). - Research and Development (R&D) expenses. - Salaries for the engineering and ROC monitoring workforce. - Local server maintenance and network infrastructure deployment costs.			REVENUE STREAMS - CAPEX (Capital Expenditure): Upfront sales of hardware retrofit packages (sensors, IPC, network nodes) and installation fees per unit. - OPEX (Operational Expenditure): Monthly Autonomous Software as a Service (SaaS) licensing fees per unit. - Annual maintenance and system calibration contracts.	

Appendix

Risk Mitigation



Risk Name	Cause	Effect	Likelihood	Impact	Score	Risk Recommendation	Mitigation / Action Items
Sensor Degradation (LiDAR/Camera)	Heavy dust, mud, rain, and severe vibrations in the pit area obscuring or damaging lenses and sensors.	AI blind spots (YOLO failure), obstacle detection failure, increasing collision risks during teleoperation.	4	3	12	Mitigate	Install auto-cleaning systems (air-blast/wipers) on LiDAR & Cameras; use IP67/IP69K ruggedized housings; implement sensor fusion (LiDAR + Radar + Camera) for redundancy.
Mesh Network & Connectivity Loss	Steep mine topography, communication dead zones, or node failures in the swarm/mesh router network.	High latency or loss of control during remote operation; delayed real-time monitoring and safety alerts.	3	3	9	Mitigate	Implement self-healing mesh protocols; add local edge fallback (auto-safe-stop if connection drops); provide a backup private LTE/5G network.
Edge AI Hardware Overheating/Failure	Extreme environmental heat, 24/7 continuous operation, and dust clogging the ventilation of rugged PCs (e.g., Jetson/Cincoze).	System downtime, AI performance throttling, loss of real-time dashboard visibility and safety alerts.	2	3	6	Mitigate	Use industrial enclosures with active/passive cooling; install thermal sensors for auto-alerts; design plug-and-play hardware kits for easy hot-swapping with spare units.
GNSS/RTK Signal Loss (Multipath Error)	Deep pit walls blocking satellite signals or signal reflection causing RTK accuracy to degrade (>15cm).	Failure of high-precision machine guidance, inaccurate geo-fencing, risk of equipment leaving safe routes.	3	2	6	Mitigate	Integrate IMU/Gyro (Sensor Fusion) and wheel odometry for Dead Reckoning during GNSS outages; deploy additional local RTK base stations in the deepest pit areas.
Teleoperation Latency / Lag	Bandwidth congestion, limited frequency spectrum, or delay in video streaming (CCTV/LiDAR point cloud).	Delayed reaction time for remote operators, potentially causing accidents or equipment damage.	2	3	6	Mitigate	Optimize video encoding (H.265) for low latency; deploy predictive AI driving assistance; enforce strict latency SLA (<100ms) with an auto-pause system if limits are breached.
Cybersecurity & Unauthorized Access	IoT devices and remote control interfaces exposed to external networks or lacking local encryption.	Hackers illegally taking control of heavy equipment, theft of operational data, or system sabotage.	1	4	4	Mitigate	Implement end-to-end encryption (TLS/SSL) for all transmissions; use secure boot and TPM modules on hardware; enforce Role-Based Access Control (RBAC) and conduct regular penetration testing.
Legacy System Integration Failure	Incompatible data formats or APIs with the client's existing Fleet Management System (FMS) or ERP.	Data silos, failure of the "plug-and-play" feature, inability to integrate the dashboard with the client's ecosystem.	2	2	4	Mitigate	Develop a flexible middleware/API gateway; conduct early technical discovery with the client's IT team; utilize industry-standard protocols (MQTT, REST API, CAN bus J1939).

Live Prototype MVP

<https://armor-pamapersada-nusantara.vercel.app/>

Link GitHub Frontend

<https://github.com/aarieffawwaz/astranauts>

Link GitHub Backend

<https://github.com/lutfialvarop/opscenter-backend>